CS 6515: Graduate Algorithms

Lecture 12: Convexity and Gradient Descent

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From linear to nonlinear optimization. An LP optimizes a linear function over a convex feasible region. We now allow a nonlinear objective, while retaining the geometric property that will make the problem tractable: convexity.

1 The offline optimization problem

We are given one fixed function $f : \mathbb{R}^d \to \mathbb{R}$ and a nonempty convex set $K \subseteq \mathbb{R}^d$. Our goal is

$$\min_{x \in K} f(x).$$

Whenever we write x^*, we assume a minimizer exists. An ε-optimal feasible answer satisfies

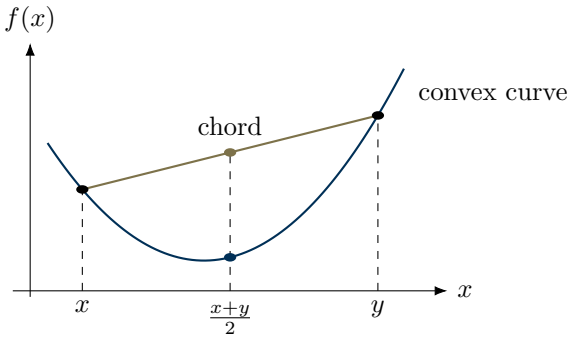
$$f(x) - f(x^*) \leq \varepsilon.$$

The input and objective are fixed throughout the computation. A value/gradient oracle lets us evaluate $f(x)$ and $\nabla f(x)$ at points we choose.

Definition (convex function). A function f is convex on a convex set K if, for all $x, y \in K$ and $\lambda \in [0, 1]$,

$$\boxed{f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y).}$$

The graph lies below its chords.



At the midpoint, $f((x+y)/2) \leq (f(x) + f(y))/2$.

A linear function is both convex and concave. Thus maximizing $c^\top x$ over an LP region is equivalent to minimizing the convex function $-c^\top x$.

2 The first-order view: tangent planes are lower bounds

For vectors, write

$$\langle u, v\rangle = u^\top v = \sum_i u_i v_i, \qquad \|u\|_2 = \sqrt{\sum_i u_i^2}, \qquad \nabla f(x) = \left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_d}\right).$$

We use the Euclidean norm throughout these two lectures.

Theorem (first-order characterization). A differentiable function $f : \mathbb{R}^d \to \mathbb{R}$ is convex if and only if, for all $x, y \in \mathbb{R}^d$,

$$\boxed{f(y) \geq f(x) + \langle \nabla f(x), y - x\rangle.}$$

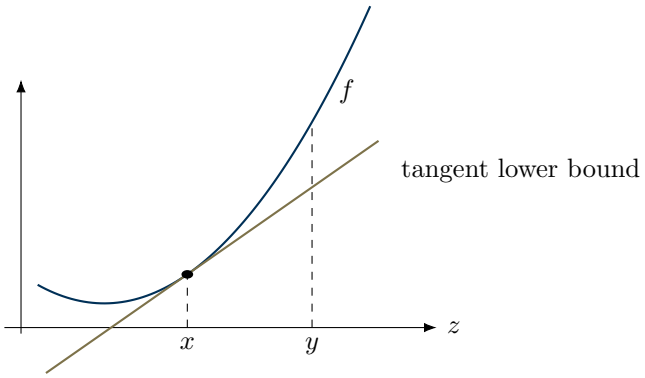
Proof of the direction we will use. For $0 < \theta \leq 1$, convexity gives

$$f(x + \theta(y - x)) \leq (1 - \theta) f(x) + \theta f(y).$$

Rearrange and let $\theta \downarrow 0$:

$$f(y) - f(x) \geq \lim_{\theta \downarrow 0} \frac{f(x + \theta(y - x)) - f(x)}{\theta} = \langle \nabla f(x), y - x\rangle.$$

For the converse, apply the first-order inequality at $z = \lambda x + (1 - \lambda)y$, once to x and once to y. Multiply by λ and $1 - \lambda$ and add: the gradient terms cancel.



The inequality to remember. Putting $y = x^*$ and rearranging,

$$\boxed{f(x) - f(x^*) \leq \langle \nabla f(x), x - x^*\rangle.}$$

This converts objective error into an inner product that we can analyze using the algorithm's update.

3 Examples and why local minima suffice

3.1 A calculus test, when second derivatives exist

For a twice continuously differentiable $f : \mathbb{R}^d \to \mathbb{R}$, convexity is equivalent to

$$\nabla^2 f(x) \succeq 0 \quad \text{for every } x.$$

Here $\nabla^2 f$ is the Hessian matrix, and positive semidefinite means $v^\top \nabla^2 f(x) v \geq 0$ for every vector v. In one dimension this is simply $f''(x) \geq 0$. Restricting f to each line gives the multivariable version of this test.

Example. For least squares,

$$f(x) = \tfrac{1}{2}\|Ax - b\|_2^2, \qquad \nabla f(x) = A^\top (Ax - b), \qquad \nabla^2 f(x) = A^\top A.$$

Since $v^\top A^\top A v = \|Av\|_2^2 \geq 0$, least squares is convex. The transpose $A^\top$ in the gradient is essential.

Example. The function $f(x) = |x|$ is convex but not differentiable at 0. A maximum of affine functions, $f(x) = \max_j (a_j^\top x + b_j)$, is also convex. Thus differentiability is an additional assumption, not part of the definition of convexity.

Theorem (local minimum implies global minimum). If f and K are convex, every local minimum of f on K is a global minimum.

Proof. Suppose $x \in K$ is a local minimum but some $y \in K$ has $f(y) < f(x)$. For every $0 < \theta \leq 1$, the feasible point $z_\theta = (1 - \theta)x + \theta y$ satisfies

$$f(z_\theta) \leq (1 - \theta) f(x) + \theta f(y) < f(x).$$

As $\theta \downarrow 0$, these better feasible points approach x, contradicting local optimality.

Stationary points and constraints. For differentiable unconstrained convex minimization, $\nabla f(x) = 0$ certifies a global minimum by the first-order inequality. The converse also holds for an unconstrained minimum. At a constrained boundary optimum the gradient need not vanish: $\min_{x \in [0,1]} x$ is optimized at 0, with derivative 1.

4 Gradient descent

First take $K = \mathbb{R}^d$. Starting at x_0, repeat

$$\boxed{x_{t+1} = x_t - \eta \nabla f(x_t), \qquad t = 0, 1, \ldots.}$$

The parameter $\eta > 0$ is the step size.

4.1 Why move in the negative-gradient direction?

For a small step h,

$$f(x + h) = f(x) + \langle \nabla f(x), h \rangle + o(\|h\|_2).$$

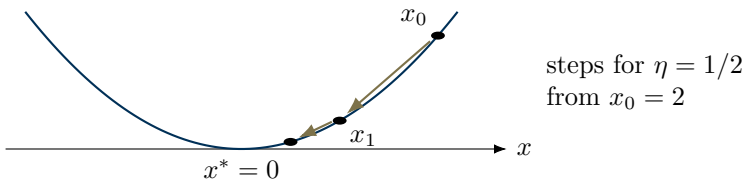
Among vectors with $\|h\|_2 \le \rho$, Cauchy–Schwarz gives

$$\langle \nabla f(x), h \rangle \ge -\rho \|\nabla f(x)\|_2.$$

If the gradient is nonzero, equality is attained in direction $-\nabla f(x)$. This is the steepest first-order decrease in Euclidean distance. The direction is useful, but the step length still matters.

Example. For $f(x) = x^2/2$, the update is $x_{t+1} = (1 - \eta)x_t$.

Step size	Behavior from $x_0 \neq 0$	Lesson
$\eta = 1/2$	$x_t = 2^{-t} x_0$	Converges
$\eta = 1$	$x_1 = 0$	Reaches the optimum immediately
$\eta = 2$	$x_t = (-1)^t x_0$	Oscillates without improving
$\eta > 2$	$\lvert x_t \rvert$ grows	Can diverge



A convergence theorem must specify assumptions, step sizes, *and which iterate or average is returned.*

5 The basic convergence guarantee

Assume $f : \mathbb{R}^d \to \mathbb{R}$ is differentiable and convex, and has a minimizer x^*. Suppose

$$\|x_0 - x^*\|_2 \le D, \qquad \|\nabla f(x_t)\|_2 \le G \quad \text{at every iterate used.}$$

Here $D, G > 0$ are known bounds used to choose the step size. Fix an iteration count T, use the same η for all steps, and return the average of the points at which gradients were queried:

$$\bar{x}_T = \frac{1}{T}\sum_{t=0}^{T-1} x_t.$$

Theorem (gradient descent with bounded gradients). For every constant step size $\eta > 0$ satisfying the assumptions above,

$$f(\bar{x}_T) - f(x^*) \le \frac{D^2}{2\eta T} + \frac{\eta G^2}{2}.$$

Choosing $\eta = D/(G\sqrt{T})$ gives

$$\boxed{f(\bar{x}_T) - f(x^*) \le \frac{DG}{\sqrt{T}}.}$$

Thus $T \ge (DG/\varepsilon)^2$ suffices for additive error ε.

5.1 Proof plan before the algebra

We cannot evaluate x^*, but we can use it in the analysis. Track the squared distance

$$R_t = \|x_t - x^*\|_2^2.$$

The update expands R_{t+1} into R_t, a gradient-squared term, and an inner product. Convexity turns that inner product into $f(x_t) - f(x^*)$.

Our target one-step inequality is

$$\boxed{f(x_t) - f(x^*) \le \frac{R_t - R_{t+1}}{2\eta} + \frac{\eta G^2}{2}.}$$

Summing makes the distance terms telescope. Convexity then turns a bound on the *average function value* into a bound at the *average point*:

$$f\left(\frac{1}{T}\sum_{t=0}^{T-1} x_t\right) \le \frac{1}{T}\sum_{t=0}^{T-1} f(x_t).$$

This is the many-point version of the chord inequality.

5.2 Step 1: expand one update

Write $g_t = \nabla f(x_t)$. Then

$$\begin{aligned} R_{t+1} &= \|x_t - \eta g_t - x^*\|_2^2 \\ &= R_t - 2\eta\langle g_t, x_t - x^*\rangle + \eta^2\|g_t\|_2^2 \\ &\le R_t - 2\eta\big(f(x_t) - f(x^*)\big) + \eta^2 G^2. \end{aligned}$$

The last line uses precisely the first-order convexity inequality and the gradient bound. Rearranging proves the target one-step inequality.

5.3 Step 2: telescope

Sum over $t = 0, \dots, T-1$:

$$\begin{aligned} \sum_{t=0}^{T-1}\big(f(x_t) - f(x^*)\big) &\le \frac{1}{2\eta}\underbrace{\sum_{t=0}^{T-1}(R_t - R_{t+1})}_{R_0 - R_T} + \frac{\eta G^2 T}{2} \\ &\le \frac{D^2}{2\eta} + \frac{\eta G^2 T}{2}, \end{aligned}$$

because $R_0 \le D^2$ and $R_T \ge 0$. Dividing by T and applying convexity to $\bar{x}_T$ proves the theorem.

5.4 Step 3: balance the two terms

For $\eta = D/(G\sqrt{T})$,

$$\frac{D^2}{2\eta T} = \frac{DG}{2\sqrt{T}}, \qquad \frac{\eta G^2}{2} = \frac{DG}{2\sqrt{T}}.$$

A tiny step keeps the discretization term small but makes progress slow. A large step does the opposite. The chosen step size balances the two.

A second valid output rule. If function values are available, choose $\hat{t} \in \arg\min_{0 \le t < T} f(x_t)$ and return $\hat{x} = x_{\hat{t}}$. This best queried iterate satisfies the same bound, since its value is at most the average value. This argument does *not* establish the same bound for the last iterate.

6 Scope, caveats, and what comes next

This is an additive approximation. The guarantee is $f(x) - f(x^*) \le \varepsilon$, not a multiplicative approximation ratio. The latter can be inappropriate when $f(x^*) = 0$ or is negative.

Count the work per iteration. The theorem bounds gradient-oracle calls. Total time also includes the cost of a gradient evaluation and vector arithmetic. For dense least squares with $A \in \mathbb{R}^{m \times d}$, computing $A^\top(Ax - b)$ takes $\mathcal{O}(md)$ time.

The bounds are real assumptions. Convexity alone does not bound D or G. For example, the gradient of $x^2/2$ is not bounded on all of $\mathbb{R}$. It is enough to bound gradients at the iterates, but the bound must be justified for the problem and step-size choice.

A short extension to nonsmooth functions. A subgradient g at x satisfies

$$f(y) \ge f(x) + \langle g, y - x \rangle \quad \text{for every } y.$$

Replacing gradients by subgradients with $\|g_t\|_2 \le G$ leaves the proof unchanged. For $f(x) = |x|$, use $g = 1$ for $x > 0$, $g = -1$ for $x < 0$, and any $g \in [-1, 1]$ at 0. This is a brief extension, not a new convergence proof.

Why projection and smoothness are next.

- A gradient step may leave a constrained feasible set K. We will project it back and preserve the distance analysis.
- A bounded gradient gives a $1/\sqrt{T}$ error bound. An upper bound on how rapidly the gradient changes will yield a faster $1/T$ guarantee.
- Lower as well as upper curvature bounds will give geometric convergence.

Exercise. With $D = 2$, $G = 3$, and $T = 900$, find the prescribed step size and the guaranteed additive error.

Exercise. Show that the average of T points in a convex set lies in that set. Then prove the many-point convexity inequality used in the analysis.

Exercise. For $f(x) = x^2/2$, check the potential calculation directly when $\eta = 1/2$. Which inequality in the general proof loses information specific to this quadratic?